Abhijeet Anand, PhD

Independent Postdoctoral Fellow

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Research Summary

I study galaxy formation and evolution through the circumgalactic and intergalactic medium, with particular emphasis on quasar absorption-line systems and the cosmic baryon (and metal) cycle. My research combines large spectroscopic surveys such as DESI, SDSS, 4MOST and WEAVE with scalable computational methods to investigate galactic gas flows, cosmic baryon cycle, and the evolution of metals across cosmic time. I also use cosmological simulations, including Illustris, TNG-100, TNG-50 and others, to connect observations with the underlying physical processes that regulate galaxy growth. In parallel, I develop community software for robust analysis of current and future spectroscopic survey data.

Professional Experience

Dec 2025 - present: Independent Postdoctoral Fellow, IUCAA, Pune, India; *Affiliate Postdoctoral Scientist*, Lawrence Berkeley National Lab, Berkeley, CA, USA (from Feb 2026)

- Leading an independent research program on the statistical properties of metals in the Universe and the physical state of gas in and around galaxies, using observations from SDSS, DESI, MUSE and HST.
- Developing community software for scalable analysis of large spectroscopic survey data, including qsoabsfind for metal absorber detection and nmfqsofit for fast, robust quasar continuum fitting.
- DESI Builder and core contributor to the DESI redshift-fitting pipeline redrock.

Sep 2022 - Oct 2025: Postdoctoral Fellow, Lawrence Berkeley National Lab, Berkeley, CA, USA

- Led independent projects on metal absorbers in and around galaxies using DESI, constructing one of the largest absorber catalogs from quasar spectra and deriving some of the tightest constraints to date on the cosmic mass density of C IV absorbers and its evolution with redshift.
- Developer and support scientist for the DESI spectral reduction pipeline (desispec); developed a hybrid archetype+PCA method in redrock for fitting galaxy spectra with synthetic templates, improving redshift measurements and reducing catastrophic failures by $\sim 30\%$.
- Contributed to DESI collaboration papers and initiated and reviewed independent science projects within and beyond the collaboration.

Sep 2018 - Jul 2022: IMPRS[1] PhD Fellow, Max Planck Institute for Astrophysics, Garching, Germany

- Studied the origin and distribution of cool metal-enriched gas in low-redshift galaxy clusters, and compared its properties with gas around field galaxies using SDSS spectroscopy and Dark Energy Survey Legacy Imaging data.
- Characterized the abundance, distribution, and nature of circumgalactic gas around star-forming and passive galaxies using SDSS quasar absorption-line systems and galaxy samples.
- Developed the first version of qsoabsfind, a tool for detecting and analyzing metal absorbers in low-resolution quasar spectra.

[1] International Max Planck Research School on Astrophysics

Sep 2017 - Jul 2018: UGC - JRF[2], National Institute of Advanced Studies, Bangalore, India

Education

Sep 2018 - Jul 2022: PhD in Astrophysics, Ludwig Maximilian University – Max Planck Institute for Astrophysics (MPA), Germany (*magna cum laude*)

- Thesis: Probing cool and warm circumgalactic gas in galaxies and clusters with large spectroscopic and imaging surveys
- Advisors: Prof. Dr. Guinevere Kauffmann & Dr. Dylan Nelson

Jul 2016 - Jul 2017: MS, Physics, Indian Institute of Science (IISc), Bangalore, India (*Grade: 7.0/8*).

- Thesis: A sensitive search for HI 21 cm emission from super disks in radio galaxies
- Advisor: Prof. Nirupam Roy

Aug 2012 - June 2016: BSc (Research), Physics, Indian Institute of Science, Bangalore, India (*Grade: 6.6/8*).

- Thesis: Sources of Continuum Opacity in Hydrogen deficient stars
- Advisors: Prof. Gajendra Pandey, Indian Institute of Astrophysics (IIA)

Selected Invited Talks and Research Visits

Sep 2026: *Department of Astronomy (DAASE) Seminar*, IIT Indore, India — invited seminar

Sep 2026: *Bridging Observations and Simulations in CGM-IGM Science*, CUHP, Dharamshala, India — invited talk

May 2026: *Department of Physics Seminar*, Presidency University, Kolkata, India — invited seminar

Apr 2026: *Galaxy Formation and Evolution Group*, Indian Institute of Astrophysics, Bangalore, India — invited research visit

June 2025: *Astro Webinar*, Raman Research Institute, Bangalore, India — invited virtual seminar

May 2025: *Astro Lunch Seminar*, University of Washington, Seattle, USA — invited seminar

Dec 2024: *Joint Astronomy Program Seminar*, IISc, Bangalore, India — invited seminar

Feb 2024: *Extragalactic Seminar*, Max Planck Institute for Astrophysics, Garching, Germany — invited seminar

Feb 2024: *Cosmology Seminar*, Max Planck Institute for Astrophysics, Garching, Germany — invited seminar

Mar 2022: *Computational Galaxy Formation and Evolution Group*, ZAH/ITA, University of Heidelberg, Heidelberg, Germany — research visit

Feb 2022: *High Energy Group Seminar*, Max Planck Institute for Extraterrestrial Physics, Garching, Germany — invited virtual seminar

Feb 2022: *DESI Group Seminar*, Lawrence Berkeley National Laboratory, Berkeley, USA — invited virtual seminar

Dec 2021: *Galaxies & Cosmology Seminar*, CfA Harvard, Cambridge, USA — invited virtual seminar

Sep 2021: *STARs Lab*, Arizona State University, USA — invited virtual seminar

Selected Contributed Talks

May 2026: *ASI Annual Meeting*, IIT Guwahati, India — contributed talk

Dec 2024: *Baryons Beyond Galactic Boundaries*, IUCAA, Pune, India — contributed talk

Nov 2024: *ISSAC Conference*, St. Stephen's College, New Delhi, India — contributed talk

[2]University Grants Commission, Govt. of India - Junior Research Fellowship

Honors and Awards

Max Planck-India Mobility Grant, Max Planck Society (2026–2029; EUR 5000 p.a.): awarded to support annual research visits to German research institutes.

IUCAA Independent Postdoctoral Fellowship (2025–2027).

DESI Builder Award (2025) for outstanding contributions to DESI data systems and survey operations.

Member of the DESI Data Systems Team, recognized as part of the DESI Collaboration with the 2026 Berkeley Prize of the American Astronomical Society.

DESI Early Career Travel Grant (2024–2025).

IMPRS PhD Fellowship (2018–2022).

University Grants Commission Junior Research Fellowship, Government of India (2017–2018).

Department of Science & Technology Higher Studies Scholarship, Government of India (2012–2017).

Mentoring and Supervision

Suman Sourav Biswal (MS, IISER Berhampur): Co-mentored with Prof. Sowgat Muzahid on a project characterizing neutral gas in the ISM of nearby galaxies (Jun–Aug 2026; IUCAA VSP student).

Roshna V (MSc, Central University of Andhra Pradesh): Co-mentored with Prof. Sowgat Muzahid on a data-driven approach to modeling quasar continua (Jun–Aug 2026; IUCAA VSP student).

Paryag Sharma (PhD, Central University of Himachal Pradesh): Co-mentoring with Prof. Raghunathan Srianand and Prof. Hum Chand on the circumgalactic medium of quasars using SDSS and DESI data (2026–present).

Sumukha Bharadwaj (PhD, Indian Institute of Space Science and Technology): Co-mentoring with Prof. Anand Narayanan on the circumgalactic medium of low-redshift galaxies using SDSS, DESI, and HST data (2026–present).

Joanne Tan (PhD, Max Planck Institute for Astrophysics): Co-mentored with Dr. Thorsten Naab on metal absorbers in the TNG100 simulation (2025–2026; now a postdoctoral researcher at ASIAA, Taiwan).

Corey Dodeson (Undergraduate, UC Berkeley): Supervised a short-term reading project on the circumgalactic medium (2023).

Software, Computation, and Survey Infrastructure

Scientific Software Leadership —
- qsoabsfind: metal absorber detection framework for low-resolution quasar spectra (*sole developer*)
- nmfqsofit: fast, robust quasar continuum-fitting framework for large spectroscopic surveys (*sole developer*)

Survey Pipeline Contributions —
- redrock: major contributor to the DESI redshift-fitting pipeline
- desispec: contributor to the DESI spectroscopic reduction pipeline

Computational Tools and Workflows —
- Python (numpy, scipy, astropy, scikit-learn, matplotlib), bash; scalable analysis of large spectroscopic datasets and high-performance computing workflows
- Git, Jupyter, Slurm (NERSC), LaTeX, Unix/Linux, CI/CD workflows

Professional Service and Academic Activities

Refereeing —
- Nature Astronomy (2026–present)
- Astronomy & Astrophysics (2023–present)

– Astrophysical Journal (2023–present)
– Internal reviewer for DESI collaboration papers (2023–present)
Leadership and Committee Service —
– Core committee member, IUCAA Computing Facilities (2026–2027)
– Core committee member, DESI Professional Development Committee (2024–2025); contributed to early-career support and mentoring initiatives
– Organizer, INPA Weekly Seminar, Physics Division, Lawrence Berkeley National Laboratory (2023–2024)
Collaboration and Community Activities —
– Continuing participant, DESI Collaboration (2026–present, membership category recognizing substantial prior contributions)
– Co-Investigator, NANCY[3] Survey on the Roman Space Telescope (2024–present)

Teaching Experience

Online physics tutor, E-acharya[4]; taught more than 20 students from June 2017 to Aug 2018.

Outreach and Public Engagement

Press and Research Highlights —
– Research featured in monthly highlights of the Max Planck Institute for Astrophysics (I, II) and on Astrobites
Interviews and Public Communication —
– Interviewed on the Vidpeds podcast about my journey in astronomy and pathways for students from small towns in India
– Featured in an interview with The Interview Portal on careers in science and astronomy
Public Presentations —
– Presented pedagogical and scientific posters for the general public on National Science Day 2026, IUCAA, Pune, India

Last update on: Sep 13, 2026

[3]Next-generation All-sky Near-infrared Community surveY
[4]An initiative to help students in suburban and rural areas of Bihar, India.